V2X Message Multi-label Classification: A Deep Learning Approach

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2025 1st International Conference on Artificial Intelligence Technology (ICoAIT) | 979-8-3315-8679-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICoAIT67446.2025.11308890

***Abstract*— This is a simulation study created to provide insights into using deep learning models to improve Vehicle-to-Everything (V2X) communications security by classifying the types of messages transmitted or received. The dataset is provided in the IEEE Data Port, pre-processed by one-hot encoding, tokenizer, post-padding, and embedding layer. After text preprocessing, the dense vectorize values are inputted into these four deep learning algorithms: Bidirectional Recurrent Neural Network (BRNN), Convolutional – Bidirectional Recurrent Neural Network (CNN-BRNN), Bidirectional Long Short-Term Memory (BLSTM), and Convolutional Bidirectional Long Short-Term Memory (CNN-BLSTM). Researchers use the TensorFlow and Keras python libraries using their default parameters of these models. A flatten and dense layer with a Softmax activation function is also used in its output layer for multilabel classification. The results of the classification metrics observed that the CNN-BRNN produced a superior performance than the rest of the models using the V2X Dataset, having an accuracy of 90%, precision of 98.46%, recall of 80.12%, f1-score of 84.49%, and loss value of 0.45. It has also been observed that BRNN has a better performance than BLSTM models with or without Convolutional Layers.**

Keywords*— *V2X, Deep Learning Models, Bidirectional Recurrent Neural Networks, Convolutional – Bidirectional Recurrent Neural Networks, Bidirectional Long Short-Term Memory, Convolutional – Bidirectional Long Short-Term Memory

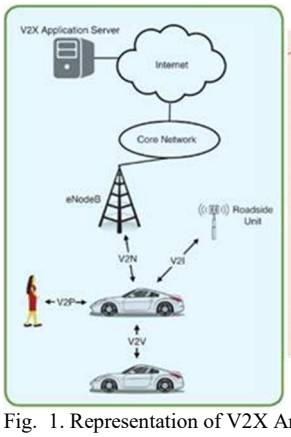
I. INTRODUCTION

There has been an increasing interest in vehicular network architectures, protocols, and applications since 2017. The core concept for all these developments is safety in all categories (vehicle, driver, passenger, pedestrian, and infrastructure). According to [1], the increasing number of people owning vehicles positively correlates to increasing vehicular accidents. Researchers are finding ways to improve the safety of owning and driving vehicles, joining with the advancement of the Internet of Things (IoT), Artificial Intelligence, and Automation. As a result, Vehicle- To-Everything (V2X) communications was introduced. The V2X communication is one of the components of the Intelligent Transportation System (ITS), which pertains to the coordination and communication of vehicles with their respective environment [2]. This communication hopes to provide improvement in economical, efficient, and safe autonomous overland transportation.

A. V2X Ecosystem

The traditional architecture and communication modes of autonomous and non-autonomous vehicles can be seen in Fig. 1 [3]. The basic architecture of V2X communications is composed of the following:

a. On-Board-Unit with Vehicle-to-Everything Communications – modern vehicles that are geared with computational power for the purpose of processing collected data (IOT / IOV) and have capabilities in transmitting and receiving information with other parts of V2X communication equipment (V2N, V2I, V2V, V2P).
b. Road Side Units – these units are selected in predetermined locations and implemented at the roadside, such as parking and intersections. These units typically use V2I communication mode.
c. Roadside Users – these include pedestrians walking in the streets, cyclists, and motorcyclists, who are prone to high-risk injury. V2P communication modes are used by the OBU.
d. Base Stations – these are components of cellular infrastructure that have V2N Communication Mode. Some studies in the deployment phase are using 5G connectivity to test the effectiveness of disseminating information in vehicles.
e. Edge or Central Cloud Servers – these are used to combine the information received from multiple sources at a designated location. These servers can contain all information that can be transmitted and received in V2X communication.



V2X Communication Modes

Mode	Description	Applications
V2N Communication	Communication between a vehicle and a V2X application server (typically using a cellular network)	Infotainment, traffic optimization, traffic safety [14], [15]
V2I Communication	Communication between a vehicle and roadside infrastructure such as road-side units (RSUs)	Traffic safety, information sharing [16], [17]
V2V Communication	Direct communication between two vehicles	Collision warning and avoidance, traffic safety [18], [19]
V2P Communication	Direct communication between vehicles and vulnerable road users (VRUs)	Traffic safety [20], [21]

Fig. 1. Representation of V2X Architecture and Communication Modes [3]

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The main purpose of the V2X technology is to integrate the vehicle with significant real-time information on displacement coordinates, speed, direction, and other emergency situations that must be considered for safe driving. However, due to different communication modes, new vulnerabilities emerged in the security infrastructure. In this study, the researchers will give insights into the possible solutions in the security infrastructure of V2X Communication in classifying messages that are received or transmitted in different nodes.

B. Security of Vehicular Networks

Security threats in V2X Communications can be classified into five categories as shown in Fig. 2. These main categories are further sub-classified to the type of attacks that have been evident in the existing works [2].

a. Authentication – these attacks can affect the process of identification and authentication synonymously, causing damage to the V2X network.
b. Availability – these attacks ensure that the V2X network is non-functional and the services of information are not available to the users at any time.
c. Confidentiality – these attacks cause violations involving the disclosure of highly private and strict information to unauthorized users.
d. Integrity – aims to manipulate the exchange of transmitted or received information/data.
e. Privacy – these attacks cause violations or threats wherein the attacker exposes the privacy of drivers to other entities or users.

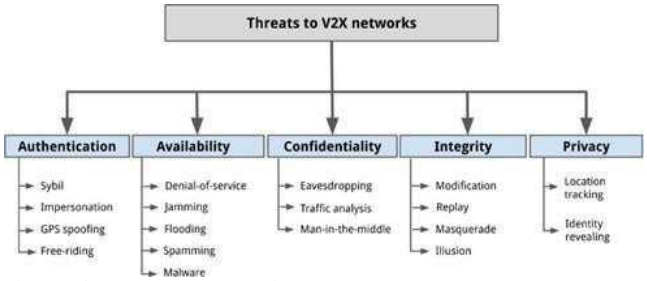


Fig. 2. Threats to V2X Networks [2]

This research will benefit the availability category in the multilabel classification of messages in the spam attack. In the Spam attack, non-priority messages are flooded by the attacker into the V2X network, which causes resource collisions due to excessive bandwidth consumption. It has been discovered that as of 2023, there are no studies, practical examples, or implementations that tackle the spamming attack activity in V2X communications [2, 4]. According to [3, 4], supervised machine learning algorithms can be provided to perform in the availability attack category so that early detection of any anonymous behavior can be identified and quickly investigated.

However, [4] emphasized that recent studies in deep learning algorithms have shown better performance over traditional supervised machine learning techniques.

The researchers hope to contribute to the V2X communication field and help other future areas as well

II. Methodology

The sections are subdivided to show how the datasets were pre-processed, processed, and applied by the researchers.

A. Dataset Labels

The data set used by the researchers is shared through a Creative Commons license in the IEEE Data Port published on May 23, 2023 [6]. It is a collection of V2X (Vehicle-to-Everything) messages designed for research and development purposes. It aims to facilitate the development and evaluation of machine learning models for V2X message analysis, including classification, prioritization, and spam message detection.

The following labels are named in the original dataset:

a. Message ID – a unique identifier for each message.
b. Source Vehicle – the vehicle from which the message originates.
c. Destination Vehicle – the message's intended destination vehicle(s).
d. Message Type – the type or category of the message, such as traffic, emergency, weather, hazard, roadwork, or spam.
e. Message Content – the actual content of the message.
f. Priority – the priority level of the message, categorized as high, medium, or low.
g. Spam – a binary label indicating whether the message is classified as spam (1) or not (0).

The researchers will only use the Message Content and Message Type labels. All algorithms will have a multilabel classification result in which the message content will be classified into the following message types: Spam, Emergency, Roadwork, Weather, Hazard, and Traffic.

B. Dataset's Distribution of Data

The distribution of data that was used by the researchersare the following: Spam messages are found at 25% (236 counts), Emergency messages are 16% (149 counts), Traffic messages are 16% (151 counts), Weather messages are 16% (146 counts), Hazard messages are 14% (131 counts). Roadwork messages are 13% (125 counts), having a total of 938 data counts.

C. Deep Learning Algorithms

As no research based on the Spamming Attack is available in the context of the V2X communication dataset, the researchers gathered information from similar datasets that have applied deep learning models for text classification. Sentiment Analysis and Topic Labeling/Marking models are found to be the closest to this study.

Sentiment Analysis is an algorithm used in the process of analyzing and reasoning the subjective text, which can be further used in both binary (positive or negative emotions) and multilabel classification. Topic Labeling/Marking predicts the meaning of the input text by defining its text theme, which is mostly used in document analysis [7]. Fig. 3 shows some of the deep learning algorithms used and can be

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summarized as having at least an accuracy standard of 60 percent to 100 percent.

Model	Sentiment						News		Topic	NLI
	MR	SST-2	IMDB	Yelp.P	Yelp.F	Amz.F	20NG	AG	DBpedia	SNLI
NB [8]	-	81.80	-	-	-	-	-	-	-	-
SVM [42]	-	79.40	-	-	-	-	-	-	-	-
Tree-CRF [284]	77.30	-	-	-	-	-	-	-	-	-
RAE [60]	77.70	82.40	-	-	-	-	-	-	-	-
MV-RNN [62]	79.00	82.90	-	-	-	-	-	-	-	-
RNTN [64]	75.90	85.40	-	-	-	-	-	-	-	-
DCNN [5]		86.80	89.40	-	-	-	-	-	-	-
Paragraph-Vec [67]		87.80	92.58	-	-	-	-	-	-	-
TextCNN[17]	81.50	88.10	-	-	-	-	-	-	-	-
TextRCNN [72]	-	-	-	-	-	-	96.49	-	-	-
DAN [69]	-	86.30	89.40	-	-	-	-	-	-	-
Tree-LSTM [1]		88.00	-	-	-	-	-	-	-	-
CharCNN [93]	-	-	-	95.12	62.05	-	-	90.49	98.45	-
HAN [108]	-	-	49.40	-	-	63.60	-	-	-	-
SeqTextRCNN [7]	-	-	-	-	-	-	-	-	-	-
oh-2LSTMp [75]	-	-	94.10	97.10	67.61	-	86.68	93.43	99.16	-
LSTMN [112]	-	87.30	-	-	-	-	-	-	-	-
Multi-Task [79]	-	87.90	91.30	-	-	-	-	-	-	-
BLSTM-2DCNN [77]	82.30	89.50	-	-	-	-	**96.50**	-	-	-
TopicRNN [83]	-	-	93.72	-	-	-	-	-	-	-
DPCNN [98]	-	-	-	97.36	69.42	65.19	-	93.13	99.12	-
KPCNN [100]	83.25	-	-	-	-	-	-	88.36	-	-
RNN-Capsule [87]	**83.80**		-	-	-	-	-	-	-	-
ULMFiT [285]	-	-	95.40	97.84	71.02	-	-	94.99	99.20	-
LEAM[286]	76.95	-	-	95.31	64.09	-	81.91	92.45	99.02	-
TextCapsule [101]	82.30	86.80	-	-	-	-	-	92.60	-	-
TextGCN [155]	76.74	-	-	-	-	-	86.34	67.61	-	-
BERT-base [19]	-	93.50	95.63	98.08	70.58	61.60	-	-	-	91.00
BERT-large [19]	-	94.90	95.79	98.19	71.38	62.20	-	-	-	91.70
MT-DNN[287]	-	95.60	83.20	-	-	-	-	-	-	91.50
XLNet-Large [138]	-	96.80	**96.21**	**98.45**	**72.20**	**67.74**	-	-	-	-
XLNet [138]	-	**97.00**	-	-	-	-	-	**95.51**	**99.38**	-
RoBERTa [140]	-	96.40	-	-	-	-	-	-	-	**92.60**

Fig. 3. Collection of Accuracy in Deep Learning Algorithms [7]

Most of the algorithms shown in Fig. 3 are composed of Recurrent Neural Networks, Convolutional Neural Networks, and Neural Network Fusion models. Other studies are also supported by the knowledge that fusion or hybrid models have been seen as far superior to a simple RNN and CNN. The following algorithms that are used in this study are inspired and discussed in the following [7, 8, 9, 10, 11, 12]:

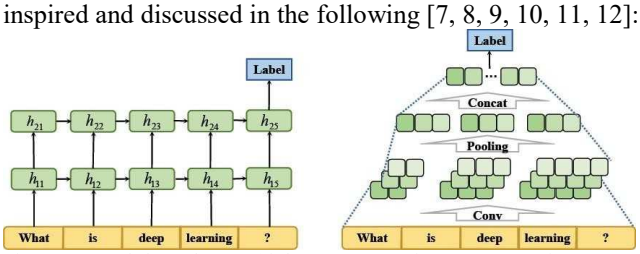


Fig. 4. RNN (left) and CNN (right) Deep Learning Representation

a. Bidirectional Recurrent Neural Networks (BRNN) – Recurrent Neural Networks are used in capturing long-range dependency by recurrent computation as it learns historical information and considers the location of each piece of information inputted for the text classification task. Fig. 4 on the left shows how RNN works. RNNs have weakness in gradient vanishing, and it is considered in the following studies that having Bidirectional Layers in Simple RNN is used to alleviate this problem.
b. Convolutional - Bidirectional Recurrent Neural Networks (CNN-BRNN) – Convolutional Neural Networks are mainly used for object detection and classifications but can also be used to produce different kernels for text sequences for the purpose of text classification. Fig. 4 on the left side shows how CNN works: a convolution layer and a pooling layer. In this study, the researchers use [8], suggesting that instead of a pooling layer to be used for filtering purposes, a Recurrent Neural Network is used. Bidirectional layers are added to the RNN to reduce the vanishing gradient problem.

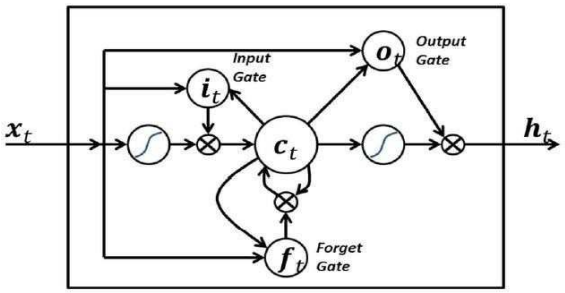


Fig. 5. LSTM Deep Learning Representation

c. Bidirectional Long Short-Term Memory Neural Networks (BLSTM) – Long Short-Term Memory is one improvised RNN Architecture used for natural language processing. It is also one of the solutions for Simple RNN problems for gradient vanishing and explosion. The proponents added a bidirectional layer in LSTM to reduce the vanishing gradient problem further. [8] Fig. 5 shows the structure of LSTM, having an input layer, forget layer and output layer.
d. Convolutional - Bidirectional Long Short-Term Memory (CNN-BLSTM) – A CNN layer is added before the BLSTM as per a similar proposal by [8]. The pooling layer is substituted by the BLSTM for more thorough long term dependency text classification

D. Process Flow Chart

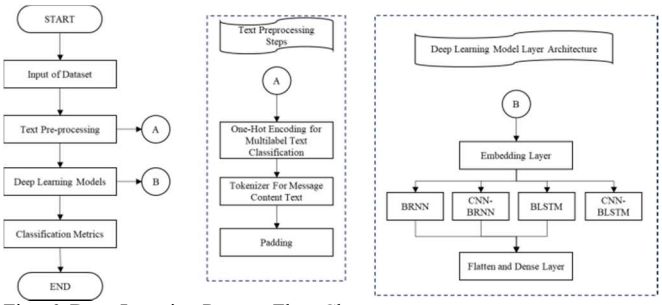


Fig. 6. Deep Learning Process Flow Chart

Fig. 6 shows the code process flow chart that the proponents used. No text feature extractions are needed in deep learning models compared with traditional machine learning algorithms. All input datasets are entered in the text preprocessing steps and directly imbued to the deep learning models. The text preprocessing and deep learning model architecture are discussed in the following:

The text pre-processing is subdivided into three processes:

a. One Hot Encoding – categorical data must be converted into vectors or tensors for the deep learning model to understand and process. As the V2X data set has six categorical classes, we assigned numerical numbers to the following: Emergency = 0, Hazard = 1, Roadwork = 2, Spam = 3, Traffic = 4, and Weather = 5.
b. Tokenizer – this class allows the vectorizing of a text corpus by turning either a sequence of integers or into a vector where the coefficient for each token could be binary, based on word count, or based on tf-idf automatically. By default, all punctuation is removed, turning all text into space- separated sequences of words. The whole message content column in the V2X data is tokenized for the deep learning model to understand.

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c. Padding – the researchers added post paddings to have a concise length of text sequences. The code is programmed only to have 12 maximum lengths, and if it is greater than 12, it is truncated.

The Deep Learning Model Layer Architecture is built on three or more layers as follows:

a. Embedding Layer – an embedding layer is used for word embedding in all models made by the researchers. Word embeddings are used to have efficient and dense representations of vectors in which similar words have similar encoding. All of the finished pre-processing steps are used in this layer to have an efficient input to each deep learning layer used.
b. Deep Learning Models - In bidirectional layers, the researchers used both Simple RNN and LSTM layers for comparisons. Fusing both bidirectional layers to a CNN layer is found to make the fusion layer more efficient and produce better performance compared to stand-alone models [8, 9]
c. Flatten and Dense Layer – Flatten layers are used in the end layer to create a single-dimensional vector for the output layer. Dense Layer is used in output layers, and in multilabel classification, softmax is the activation used. The researchers use two dense layers, with the first dense layer having sigmoid activation and following to having the second dense layer of softmax activation for multilabel classification

E. Metrics

This study used the validation split of 0.3. In regards to the performance of all algorithm models, each classification report is shown to build a report indicating the main metrics needed. These metrics are summarized in this section [7, 13]:

a. Accuracy—this function metric in a multilabel classification computed in the set of predicted labels for a sample that absolutely matches the trained set of labels. The calculation is shown in Eq. (1).

$$\text{accuracy} = (y, \hat{y}) = \frac{1}{n_{samples}} \sum_{i=0}^{n_{samples}-1} 1(\hat{y}_i = y_i) \quad (1)$$

where: $\hat{y}_i$ = predicted output values,

y_i = true value output

b. Precision—this is the ratio between the number of true positives and false positives. This is the function of the multilabel classifier to approximate any false positives. The calculation is shown in Eq. (2).

$$\text{Precision} = \frac{t_p}{t_p + f_p} \quad (2)$$

where: t_p = number of true positives,

f_p = the number of false positives

c. Recall—this is the ratio between the number of true positives and the number of false negatives. The calculation is shown in Eq. (3). This is the function of the multilabel classifier not to have any false negatives.

$$\text{Recall} = \frac{t_p}{t_p + f_n} \quad (3)$$

where: t_p = number of true positives,

f_n = the number of false negatives

d. F1-Score—this is the weighted harmonic mean of the precision and recall, as shown in Eq. (4). This is used with accuracy to determine the model's performance on a per-class basis in multilabel classification models.

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

e. Loss—the researchers used the probabilistic losses function of CategoricalCrossentropy. Categorical CrossEntropy computes the cross-entropy loss between the difference of the label value and the prediction value. This function works efficiently in multiple labeled one-hot encoding classes, which are also used by the researchers in this study for text preprocessing.

III. Result and Discussion

This section is divided into the four deep learning algorithm results by the metrics. The number of epochs is made to be equal to 10, and all default parameters are used in this model.

A. Bidirectional Recurrent Neural Network

The ten epochs performed in the BRNN model suggest high compatibility with the V2X Dataset, as shown in Table I. Upon reaching the fifth epoch, accuracy, precision, recall and F1-score already achieved the 95 to 99 percent scores. The loss value settled in the tenth epoch to 0.14.

TABLE I. BRNN Classification Metrics

Epoch	Accuracy	Loss	Precision	Recall	F1-Score
1	34.45%	1.62	97.37%	5.64%	10.66%
2	70.73%	1.02	96.28%	31.55%	47.53%
3	92.53%	0.63	99.54%	65.85%	79.27%
4	96.80%	0.45	99.26%	82.32%	90.00%
5	98.32%	0.34	99.21%	95.73%	97.44%
6	98.93%	0.27	99.54%	98.17%	98.85%
7	99.39%	0.23	99.69%	98.78%	99.23%
8	99.39%	0.19	99.85%	99.39%	99.62%
9	99.39%	0.17	99.69%	99.39%	99.54%
10	99.85%	0.14	99.85%	99.39%	99.62%

Fig. 7 shows the upward trend of the results of accuracy, precision, recall, and f1-score until the values reached a ceiling of 99 percent in the tenth epoch, while the loss values are in the downward trend of the range 0.14 – 0.34 values starting at the fifth epoch until the end.

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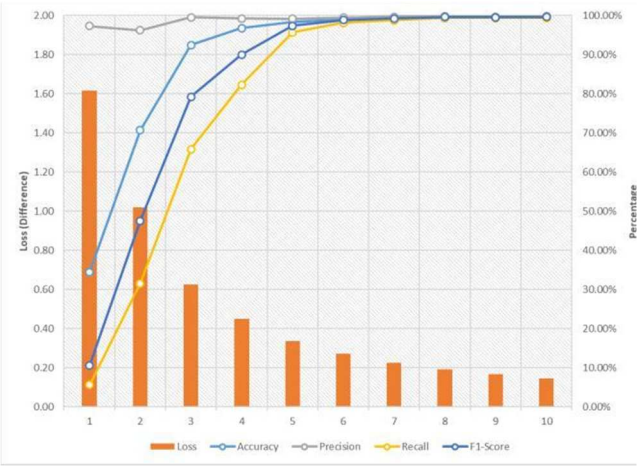


Fig. 7. BRNN Classification Metric Graph

B. CNN-BRNN

The CNN-BRNN model shows in Table II greater and improved metric values compared to having only a BRNN. The accuracy, precision, recall, and F1-Score shown in the fifth epoch achieve already 96 – 99 percent but with a smaller loss value of 0.2 in BRNN.

TABLE II. CNN-BRNN CLASSIFICATION METRICS

Epoch	Accuracy	Loss	Precision	Recall	F1-Score
1	44.51%	1.50	90.16%	8.38%	15.34%
2	71.49%	0.89	98.58%	42.38%	59.28%
3	91.46%	0.56	99.57%	70.27%	82.39%
4	97.41%	0.38	99.66%	89.02%	94.04%
5	98.63%	0.29	99.22%	96.34%	97.76%
6	98.93%	0.23	99.38%	98.32%	98.85%
7	99.24%	0.19	99.23%	98.78%	99.01%
8	99.24%	0.16	99.54%	98.78%	99.16%
9	99.54%	0.14	99.54%	99.39%	99.47%
10	99.54%	0.12	99.69%	99.54%	99.62%

Fig. 8 shows the upward trend of the accuracy, precision, recall, and F1-Score with the corresponding downward trend of the loss value. It has the lowest starting loss value in the first epoch as well in all four deep learning models.

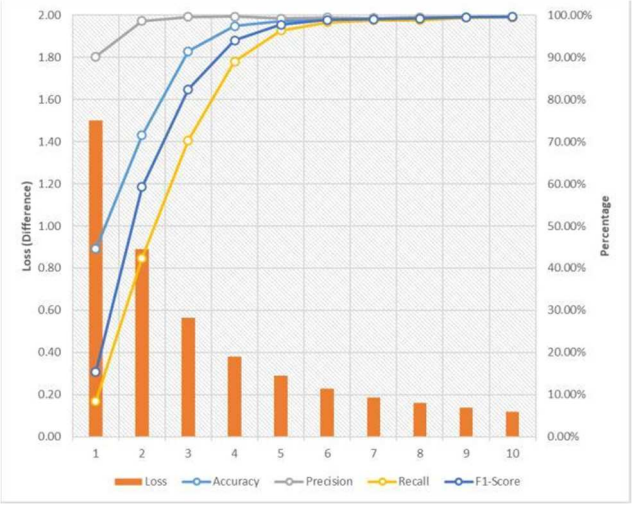


Fig. 8. CNN-BRNN Classification Metric Graph

C. Bidirectional LSTM

The classification metrics for the BLSTM shown in Table III state a lower performance compared to the BRNN. The model started to peak at 90 – 99 percent until the ninth epoch. Loss values compared to BRNN and CNN-BRNN also show a lesser performance, having only a 0.26.

TABLE III. BLSTM CLASSIFICATION METRICS

Epoch	Accuracy	Loss	Precision	Recall	F1-Score
1	19.97%	1.85	0.00%	0.00%	0.00%
2	28.66%	1.71	0.00%	0.00%	0.00%
3	54.12%	1.28	100.00%	19.51%	32.65%
4	71.95%	0.92	98.60%	32.16%	48.51%
5	80.18%	0.71	97.66%	44.51%	61.15%
6	96.49%	0.52	98.17%	81.55%	89.09%
7	97.71%	0.39	98.26%	86.13%	91.80%
8	98.32%	0.32	98.80%	87.80%	92.98%
9	98.93%	0.27	98.93%	98.48%	98.70%
10	98.02%	0.26	98.16%	97.56%	97.86%

Fig. 9 shows how the classification metrics of accuracy, recall, and F1-score struggle to have an upward trend in meeting similar results in the BRNN and CNN-BRNN until the sixth to eighth epoch values. It has also been shown to have a high precision value as it starts in the third epoch onwards. The loss values showed a slow downward trend as well compared to the first two deep learning models used.

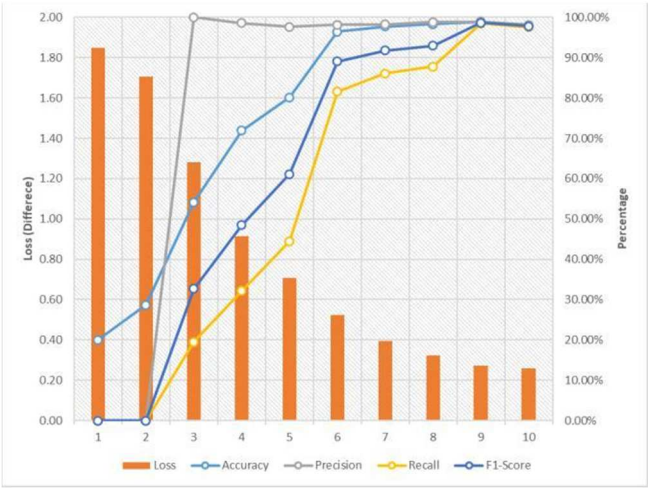


Fig. 9. BLSTM Classification Metric Graph

D. CNN-Bidirectional LSTM

Adding a CNN layer before the BLSTM has proved to improve the model for text classification. Table IV shows that it has an earlier ceiling of classification metrics values of accuracy, precision, recall, and F1-Score by the seventh epoch. The loss values compared to BLTSM show almost the same value of 0.26-0.27 in the final epoch.

Fig. 10 shows a similar struggling upward trend of classification metrics of accuracy, recall, and F1-Score but has risen two epochs earlier compared to a BLSTM. Precision also has similar results of having 90 percent at the start of the second epoch onwards. The loss values are also similar, with a BLSTM having a slow downward trend.

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TABLE IV. CNN-BLSTM CLASSIFICATION METRICS

Epoch	Accuracy	Loss	Precision	Recall	F1-Score
1	21.34%	1.82	0.00%	0.00%	0.00%
2	39.48%	1.52	92.63%	13.41%	23.44%
3	51.52%	1.18	97.38%	28.35%	43.92%
4	64.18%	0.96	95.49%	38.72%	55.10%
5	77.74%	0.77	93.77%	43.60%	59.52%
6	94.36%	0.55	99.27%	62.50%	76.71%
7	97.41%	0.40	98.43%	95.27%	96.82%
8	98.17%	0.32	98.32%	98.02%	98.17%
9	97.87%	0.30	98.15%	97.26%	97.70%
10	97.10%	0.27	97.99%	96.65%	97.31%

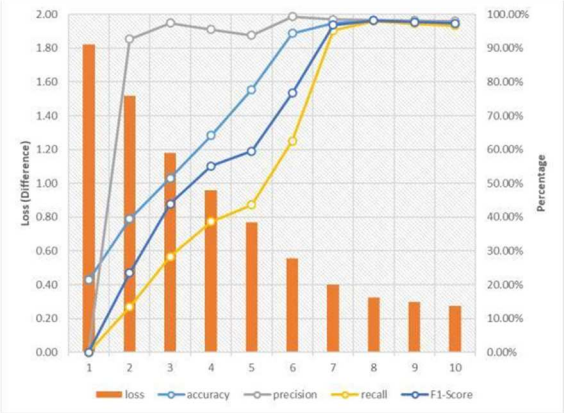


Fig. 10. CNN-BLSTM Classification Metric Graph

IV. CONCLUSION

Table V shows the means of the results for the researchers to compare all four deep learning models used. It can be deduced that the CNN-BRNN produces the highest performance, having an accuracy of 90 percent, precision of 98.46 percent, recall of 80.12 percent, f1-score of 84.49 percent, and loss value of 0.45. It has also been shown that the use of a Bidirectional Recurrent Neural Network using a Simple RNN layer has better performance than having an LSTM layer in the V2X data set. It is also proved that using CNN layer with a BRNN and BLSTM architectures as a substitute to the normal pooling layer provides a better performance than the BRNN and BLSTM itself.

TABLE V. SUMMARY OF DEEP LEARNING CLASSIFICATION METRIC

Model	Accuracy	Loss	Precision	Recall	F1-Score
BRNN	88.98	0.51	99.03	77.62	82.18
CNN-BRNN	**90.00**	**0.45**	**98.46**	**80.12**	**84.49**
BLSTM	74.44	0.82	78.86	54.77	61.27
CNN-BLSTM	73.92	0.81	87.14	57.38	64.87

ACKNOWLEDGMENT

Soli Deo Gloria. I want to thank Sai Shibu, who shared this data in the IEEE Data Port, and my professors at the Polytechnic University of the Philippines for their kind-hearted encouragement to bring this study alive. I also want to thank my family, friends, and colleagues who have supported and encouraged me to finish this paper. I hope this can help the community in the field of V2X communications. Deus benedicat.

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